RDF/XML AND N3

A LIGHT INTRODUCTION TO THEIR SYNTAX

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RDF/XML

```
<!-- First let's make it easier for ourselves by adding abbreviations -->
  <rdf:rdf xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:homemadeont="http://myBlog.com/homemade-ont/" xmlns

  <!-- Here's an RDF triple. -->
    <rdf:description rdf:about="http://someSocialNetwork.com/people#mari">
      <homemadeont:hasjob rdf:resource="work:sysadmin">
    </homemadeont:hasjob></rdf:description>

  <!-- The underlying RDF structure is there -->
    <rdf:description rdf:about="subject">
      <predicate rdf:resource="object">
    </predicate></rdf:description>
  </rdf:rdf>
```

Note the use of xml namespaces. Click **down arrow** to see translation.

N3

```
@prefix homeMadeOnt: <http://myBlog.com/homemade-ont/>.
@prefix work: <http://someOntologyBase.org/tech-positions#>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

#With N3 triples become oneliners of subject predicate object

<http://someSocialNetwork.com/people#mari> homeMadeOnt:hasJob work:sysadmin.

# Comments are made using the '#' sign
```

Click **up arrow** to go back. All the rdf/xml have corresponding N3 translations.

MULTIPLE PROPERTY ELEMENTS XML

```
<rdf:rdf xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:foaf="http://xmlns.com/foaf/0.1/" xmlns:sns="http://s

  <rdf:description rdf:about="sns:greg">
    <foaf:knows>
      <rdf:description rdf:about="sns:mina"></rdf:description>
    </foaf:knows>
    <foaf:knows>
      <rdf:description rdf:about="sns:ole"></rdf:description>
    </foaf:knows>
  </rdf:description>

</rdf:rdf>
```

MULTIPLE PROPERTY ELEMENTS N3

```
@prefix sns: <http://someSocialNetwork.com/people#>.
@prefix foaf: <http://xmlns.com/foaf/0.1/>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

sns:greg foaf:knows sns:mina,
                    sns:ole.
```

EMPTY PROPERTY ELEMENTS XML

```
<rdf:rdf xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:foaf="http://xmlns.com/foaf/0.1/" xmlns:sns="http://s

  <rdf:description rdf:about="sns:greg">
    <foaf:knows>
      <rdf:description rdf:about="sns:ole"></rdf:description>
    </foaf:knows>
    <foaf:knows rdf:resource="sns:mina"></foaf:knows> <!-- Shortened version -->
  </rdf:description>

</rdf:rdf>
```

EMPTY PROPERTY ELEMENTS N3

```
@prefix sns: <http://someSocialNetwork.com/people#>.
@prefix foaf: <http://xmlns.com/foaf/0.1/>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

sns:greg foaf:knows sns:mina,
                    sns:ole.
```

The corresponding n3 remains the same.

PROPERTY ATTRIBUTES XML

```
<rdf:rdf xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:foaf="http://xmlns.com/foaf/0.1/" xmlns:sns="http://s

  <rdf:description rdf:about="sns:greg">
    <foaf:knows rdf:resource="sns:mina">
    <foaf:name>Greg Doe</foaf:name> <!-- String literals -->
  </foaf:knows></rdf:description>
  <rdf:description rdf:about="sns:jane" foaf:name="Jane Doe"> <!-- can be shortened like so -->
    <foaf:knows rdf:resource="sns:greg">
  </foaf:knows></rdf:description>

</rdf:rdf>
```

PROPERTY ATTRIBUTES N3

```
@prefix sns: <http://someSocialNetwork.com/people#>.
@prefix foaf: <http://xmlns.com/foaf/0.1/>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

sns:greg foaf:knows sns:mina;
  foaf:name "Greg Doe".

sns:jane foaf:knows sns:greg;
  foaf:name "Jane Doe".
```

XML LITERALS

```
<rdf:rdf xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:sns="http://someSocialNetwork.com/people#" xmlns:ex="

  <rdf:description rdf:about="sns:greg">
    <ex:hasphone rdf:parsetype="Literal" xmlns:a="http://phoneXMLNamespace.org/a#">
      <a:rawcontactxml>
        <a:contact celnum="123 456 78">Jane Doe</a:contact>
      </a:rawcontactxml>
    </ex:hasphone>
  </rdf:description>

</rdf:rdf>
```

XML LITERALS N3

```
@prefix sns: <http://someSocialNetwork.com/people#>.
@prefix xmlns: <http://www.w3.org/2000/xmlns/>.
@prefix ex: <http://someExampleOntology.org/>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

sns:greg ex:hasPhone "
  <a:rawContactXML xmlns:a=\"http://phoneXMLNamespace.org/a#\">
    <a:contact celNum=\"123 456 78\">Jane Doe</a:contact>
  </a:rawContactXML>
  "^^rdf:XMLLiteral.

#The above can be written in one line
sns:greg ex:hasPhone "<a:rawContactXML xmlns:a=\"http://phoneXMLNamespace.org/a#\"> <a:contact celNum=\"123 456 78\">Jane Do
```

TYPED LITERALS RDF:DATATYPE

```
<rdf:rdf xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:sns="http://someSocialNetwork.com/people#" xmlns:foaf

  <rdf:description rdf:about="sns:greg">
    <foaf:name rdf:datatype="http://www.w3.org/2001/XMLSchema#string">Greg Doe</foaf:name>
    <foaf:age rdf:datatype="http://www.w3.org/2001/XMLSchema#int">24</foaf:age>
  </rdf:description>

</rdf:rdf>
```

N3

```
@prefix sns: <http://someSocialNetwork.com/people#>.
@prefix foaf: <http://xmlns.com/foaf/0.1/>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

sns:greg foaf:age 24;
  foaf:name "Greg Doe"^^<http://www.w3.org/2001/XMLSchema#string>.
```

IDENTIFYING BLANK NODES - RDF:NODEID

```
<rdf:rdf xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:sns="http://someSocialNetwork.com/people#" xmlns:foaf

  <rdf:description rdf:about="sns:greg">
    <foaf:knows>
      <rdf:description rdf:about="sns:john"></rdf:description>
    </foaf:knows>
    <foaf:knows rdf:nodeid="abc"></foaf:knows> <!-- We create a local reference -->
  </rdf:description>

  <rdf:description rdf:nodeid="abc"> <!-- and make a reference to it. -->
    <foaf:name>Jane Doe</foaf:name>
    <foaf:homepage>http://jane.blogg.no/</foaf:homepage>
  </rdf:description>

</rdf:rdf>
```

IDENTIFYING BLANK NODES - RDF:NODEID

```
@prefix sns: <http://someSocialNetwork.com/people#>.
@prefix foaf: <http://xmlns.com/foaf/0.1/>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

sns:greg foaf:knows sns:john,
    _:abc.

_:abc foaf:homepage "http://jane.blogg.no/";
  foaf:name "Jane Doe".
```

OMITING BLANK NODES

```
<rdf:rdf xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:sns="http://someSocialNetwork.com/people#" xmlns:foaf
  <rdf:description rdf:about="sns:greg">
    <!-- This -->
    <foaf:knows>
      <rdf:description foaf:name="Jane Doe">
        <foaf:homepage>http://jane.blogg.no/</foaf:homepage>
      </rdf:description>
    </foaf:knows>
    <!-- can be written like so, omitting the rdf:description tags -->
    <foaf:knows rdf:parsetype="Resource">
      <foaf:name>John Doe</foaf:name>
      <foaf:homepage>http://john.blogg.no/</foaf:homepage>
    </foaf:knows>
  </rdf:description>
</rdf:rdf>
```

OMITING BLANK NODES - N3

```
@prefix sns: <http://someSocialNetwork.com/people#>.
@prefix foaf: <http://xmlns.com/foaf/0.1/>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

sns:greg foaf:knows _:bnode136700096,
    _:bnode800626944.
_:bnode136700096 foaf:homepage "http://john.blogg.no/";
  foaf:name "John Doe".
_:bnode800626944 foaf:homepage "http://jane.blogg.no/";
  foaf:name "Jane Doe".
```

OMITTING NODES, CONTINUED

```
<rdf:rdf xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:sns="http://someSocialNetwork.com/people#" xmlns:foaf

  <rdf:description rdf:about="sns:greg">
    <foaf:knows rdf:parsetype="Resource">
      <foaf:name>John Doe</foaf:name>
      <foaf:homepage>http://john.blogg.no/</foaf:homepage>
    </foaf:knows>

    <!-- yet another abbreviation possibility -->
    <foaf:knows foaf:name="John Doe" foaf:homepage=" http://john.blogg.no/">

  </foaf:knows></rdf:description>

</rdf:rdf>
```

Omitting nodes: Property attributes on an empty property element

OMITTING NODES, CONTINUED - N3

```
@prefix sns: <http://someSocialNetwork.com/people#>.
@prefix foaf: <http://xmlns.com/foaf/0.1/>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

#Here Greg knows to people who *might* be the same person.

sns:greg foaf:knows _:bnode1421465472,
     _:bnode134694720.
_:bnode134694720 foaf:homepage " http://john.blogg.no/";
  foaf:name "John Doe".
_:bnode1421465472 foaf:homepage "http://john.blogg.no/";
  foaf:name "John Doe".
```

TYPED NODE ELEMENTS

```
<rdf:rdf xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:sns="http://someSocialNetwork.com/people#" xmlns:foaf

  <!-- Statements like this -->
  <rdf:description rdf:about="sns:greg">
    <rdf:type rdf:resource="foaf:Person"></rdf:type>
    <foaf:name>Greg Doe</foaf:name>
  </rdf:description>

  <!-- may also be written like this -->
  <foaf:person rdf:about="sns:jane">
    <foaf:name>Jane Doe</foaf:name>
  </foaf:person>

</rdf:rdf>
```

TYPED NODE ELEMENTS - N3

```
@prefix sns: <http://someSocialNetwork.com/people#>.
@prefix foaf: <http://xmlns.com/foaf/0.1/>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

sns:greg a foaf:Person;
  foaf:name "Greg Doe".
sns:jane a foaf:Person;
  foaf:name "Jane Doe".
```

ABBREVIATING URIS WITH XML:BASE

```
<!-- With xml:base we defined a new base URL -->
<rdf:rdf xml:base="http://newSocialNetwork.com/persons/" xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:sns="

  <!-- which allow for even more abbreviations -->
  <rdf:description rdf:id="Margareth">
    <foaf:knows rdf:resource="Morten"></foaf:knows>
    <foaf:knows rdf:resource="sns:Greg"></foaf:knows>
  </rdf:description>

</rdf:rdf>
```

These abbreviations are possible for rdf:about, rdf:resource, rdf:ID and rdf:datatype.

ABBREVIATING URIS WITH XML:BASE N3

```
@prefix sns: <http://someSocialNetwork.com/people#>.
@prefix foaf: <http://xmlns.com/foaf/0.1/>.
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

<http://newsocialnetwork.com/persons/#Margareth> foaf:knows sns:Greg,
    <http://newsocialnetwork.com/persons/Morten>.
```

OTHER RDF/XML TRICKS

- The *<rdf:RDF></rdf:RDF>* top node may be omited if the remaining rdf/xml has a single top node like *<rdf:Description></rdf:Description>*.
- *xml:lang=""* may be used on any node element or property element to indicate the language.

N3

SLIDES

Not a coder? No problem. There's a fully-featured visual editor for authoring these, try it out at **http://slid.es**

POINT OF VIEW

Press **ESC** to enter the slide overview.

Hold down alt and click on any element to zoom in on it using **zoom.js** Alt + click anywhere to zoom back out.

WORKS IN MOBILE SAFARI

Try it out! You can swipe through the slides and pinch your way to the overview.

MARVELOUS UNORDERED LIST

- No order here
- Or here
- Or here
- Or here

FANTASTIC ORDERED LIST

1. One is smaller than...
2. Two is smaller than...
3. Three!

MARKDOWN SUPPORT

For those of you who like that sort of thing. Instructions and a bit more info available **here**

```
<section data-markdown>
  ## Markdown support

  For those of you who like that sort of thing.
  Instructions and a bit more info available [here](https://github.com/hakimel/reveal.js#markdown).
</section>
```

TRANSITION STYLES

You can select from different transitions, like:
Cube - **Page** - **Concave** - **Zoom** - **Linear** - **Fade** - **None** - **Default**

THEMES

Reveal.js comes with a few themes built in:
Default **Sky** - **Beige** - **Simple** - **Serif** - **Night**
Moon - **Simple** - **Solarized**

* Theme demos are loaded after the presentation which leads to flicker. In production you should load your theme in the `<head>` using a `<link>`.

GLOBAL STATE

Set `data-state="something"` on a slide and "`something`" will be added as a class to the document element when the slide is open. This lets you apply broader style changes, like switching the background.

CUSTOM EVENTS

Additionally custom events can be triggered on a per slide basis by binding to the `data-state` name.

```
Reveal.addEventListener( 'customevent', function() {
  console.log( '"customevent" has fired' );
} );
```

SLIDE BACKGROUNDS

Set `data-background="#007777"` on a slide to change the full page background to the given color.
All CSS color formats are supported.

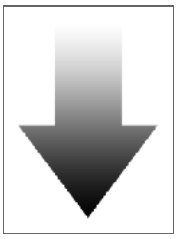


IMAGE BACKGROUNDS

```
<section data-background="image.png">
```

REPEATED IMAGE BACKGROUNDS

```
<section data-background="image.png" data-background-repeat="repeat" data-background-size="100px">
```

BACKGROUND TRANSITIONS

Pass reveal.js the `backgroundTransition: 'slide'` config argument to make backgrounds slide rather than fade.

BACKGROUND TRANSITION OVERRIDE

You can override background transitions per slide by using `data-background-transition="slide"`.

CLEVER QUOTES

These guys come in two forms, inline: *"The nice thing about standards is that there are so many to choose from"* and block:

> *"For years there has been a theory that millions of monkeys typing at random on millions of typewriters would reproduce the entire works of Shakespeare. The Internet has proven this theory to be untrue."*

PRETTY CODE

```
function linkify( selector ) {
  if( supports3DTransforms ) {

    var nodes = document.querySelectorAll( selector );

    for( var i = 0, len = nodes.length; i < len; i++ ) {
      var node = nodes[i];

      if( !node.className ) {
        node.className += ' roll';
      }
    }
  }
}
```

Courtesy of **highlight.js**

INTERGALACTIC INTERCONNECTIONS

You can link between slides internally, **like this**

FRAGMENTED VIEWS

Hit the next arrow...

... to step through ...

1. `any type`
2. *cf view*
3. **fragments**

FRAGMENT STYLES

There's a few styles of fragments, like:

grow

shrink

roll-in

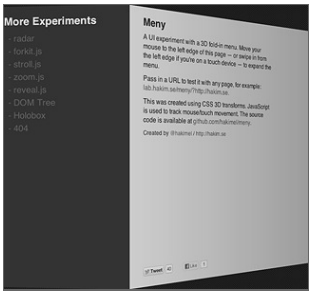
fade-out

highlight-red

highlight-green

highlight-blue

SPECTACULAR IMAGE!



EXPORT TO PDF

Presentations can be **exported to PDF**, below is an example that's been uploaded to SlideShare.

TAKE A MOMENT

Press b or period on your keyboard to enter the 'paused' mode. This mode is helpful when you want to take distracting slides off the screen during a presentation.

STELLAR LINKS

- **Try the online editor**
- **Source code on GitHub**
- **Follow me on Twitter**

THE END

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